

GNSS Sensor Deep Analysis Report

Three-Dimensional Core Analysis

Analysis Dimensions:

1. Vertical Precision Tiering
 2. Environmental Coupling Analysis
 3. Diurnal Cycle & Multipath Detection
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Dataset: sensor_data_clean_stable.csv

Devices Analyzed: 8

Total Records: 135,168

Time Range: 2025-11-10 to 2025-11-26

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1 Executive Summary

This report presents a comprehensive three-dimensional analysis of GNSS sensor performance data collected from 8 devices over a 16-day period. The analysis focuses on vertical positioning accuracy, environmental factor coupling, and temporal patterns that may indicate multipath interference.

1.1 Key Findings

- **Vertical Precision:** All 8 devices (100%) fall into Tier 3 (Warning) category with altitude STD > 5m. No devices achieved high-precision status.
- **Environmental Coupling:** Weak or anomalous altitude-pressure correlations detected across all devices, suggesting potential sensor calibration issues or environmental interference.
- **Multipath Effects:** Significant 24-hour periodic patterns observed with diurnal amplitudes ranging from 11.17m to 21.68m, indicating probable multipath interference from nearby reflective surfaces.

1.2 Recommendations

1. Investigate antenna installation environments for reflective surfaces
2. Consider implementing RTK/PPP differential corrections
3. Flag device 1878250224 for hardware inspection (903m altitude range anomaly)
4. Implement data quality filtering for outliers beyond 3σ

2 Dimension 1: Vertical Precision Tiering

2.1 Methodology

Vertical precision is assessed by calculating the standard deviation (STD) of MSL altitude measurements for each device. Devices are classified into three tiers based on precision thresholds:

Table 1: Precision Tier Definitions

Tier	Classification	Threshold	Application
Tier 1	High Precision	$STD \leq 1.5m$	Survey-grade applications
Tier 2	Standard	$STD \leq 5.0m$	Navigation & tracking
Tier 3	Warning	$STD > 5.0m$	Requires investigation

2.2 Results

Table 2: Device Precision Tier Distribution

Tier	Device Count	Percentage	Status
Tier 1 (High)	0	0.0%	—
Tier 2 (Standard)	0	0.0%	—
Tier 3 (Warning)	8	100.0%	■

Table 3: Individual Device Statistics (Sorted by Stability)

Device ID (Last 10)	Alt STD (m)	Alt Range (m)	Avg Sats	Avg HDOP	Records
...9994605977	5.01	57.6	12.0	0.64	16,925
...4811437779	7.07	96.4	12.0	0.65	19,039
...0782527426	7.95	179.1	12.0	0.63	14,185
...5642508217	9.53	254.6	12.0	0.71	15,971
...6331369164	15.21	156.9	12.0	0.81	12,913
...3827373510	15.63	229.0	12.0	0.60	17,991
...3816948226	16.45	402.0	12.0	0.73	18,462
...1878250224	18.24	903.3	12.0	1.50	19,682

Note: Device 1878250224 exhibits an extreme altitude range of 903.3m with a minimum recorded altitude of -591m, indicating severe data quality issues or hardware malfunction.

2.3 Visualization

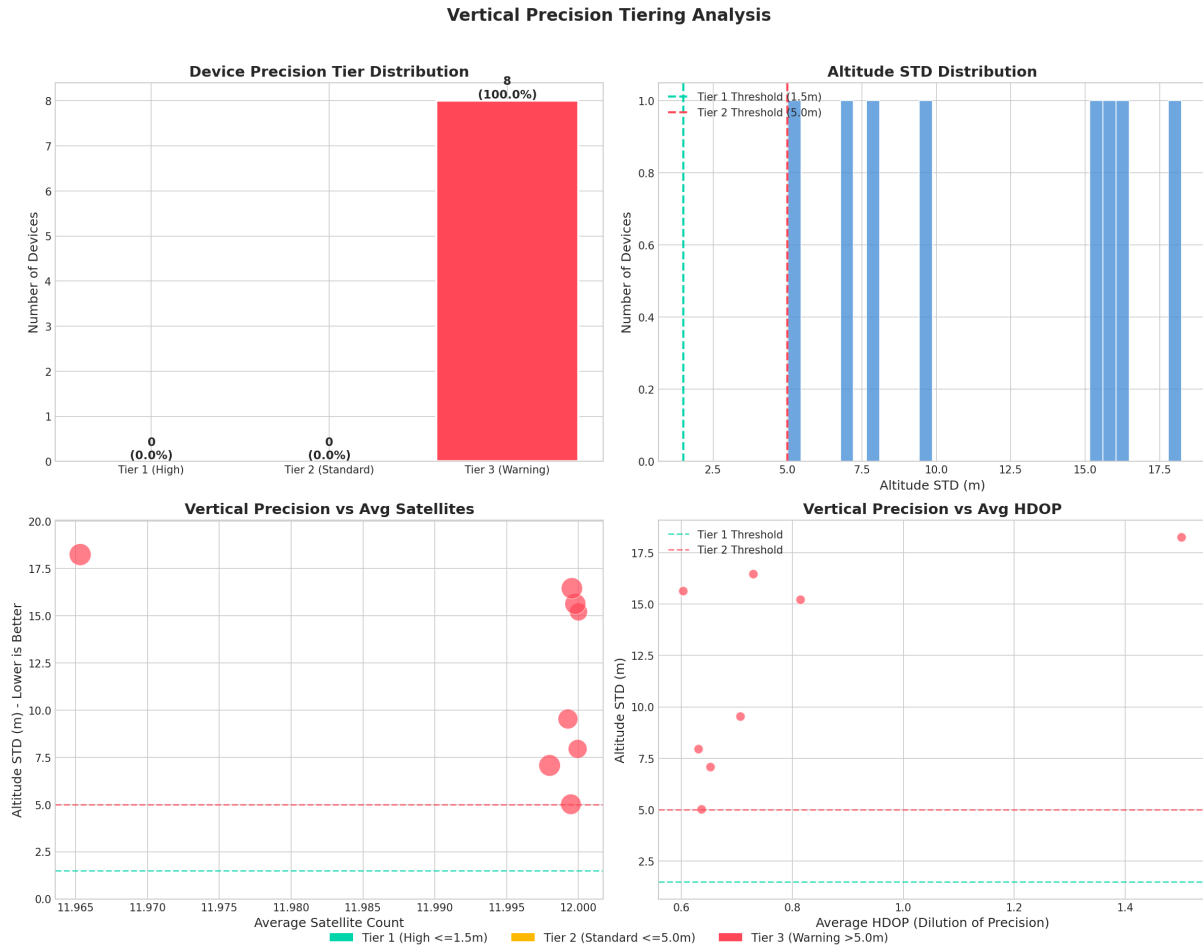


Figure 1: Vertical Precision Tiering Analysis. Top-left: Device distribution by tier. Top-right: Altitude STD histogram with tier thresholds. Bottom-left: STD vs satellite count relationship. Bottom-right: STD vs HDOP correlation.

2.4 Analysis

- All devices maintain high satellite counts (≈ 12) and low HDOP values (< 1.0 for most), yet exhibit poor vertical precision
- This decoupling between HDOP and actual precision suggests that traditional DOP metrics may not fully capture vertical accuracy in this deployment
- The most stable device (9994605977) achieves STD of 5.01m, just above the Tier 2 threshold
- Device 1878250224 shows significantly higher HDOP (1.50) correlating with worst precision

3 Dimension 2: Environmental Coupling Analysis

3.1 Theoretical Background

According to the barometric formula, atmospheric pressure P decreases with altitude h :

$$P = P_0 \cdot \exp\left(-\frac{Mgh}{RT}\right) \quad (1)$$

where P_0 is sea-level pressure, M is molar mass of air, g is gravitational acceleration, R is the gas constant, and T is temperature.

For GNSS altitude measurements to be physically consistent, we expect:

- **Strong negative correlation** ($r < -0.5$) between altitude and pressure
- Weak or no correlation between altitude and temperature/humidity (unless thermal drift exists)

3.2 Results

3.2.1 Global Correlation Matrix (All Devices)

Table 4: Environmental Factor Correlation Matrix

	Altitude	Pressure	Temperature	Humidity
Altitude	1.0000	-0.7258	-0.0309	-0.0141
Pressure	-0.7258	1.0000	-0.1553	-0.1714
Temperature	-0.0309	-0.1553	1.0000	-0.3215
Humidity	-0.0141	-0.1714	-0.3215	1.0000

Observation: Global correlation shows $r = -0.7258$ between altitude and pressure, which appears reasonable. However, this is largely driven by altitude differences *between* devices at different locations.

3.2.2 Single-Device Correlation (Reference Device)

When analyzing a single device (eliminating location variance):

Table 5: Reference Device (...78250224) Correlation

	Altitude	Pressure	Temperature	Humidity
Altitude	1.0000	+0.0465	-0.0236	-0.0562
Pressure	+0.0465	1.0000	-0.5407	-0.4611
Temperature	-0.0236	-0.5407	1.0000	0.2994
Humidity	-0.0562	-0.4611	0.2994	1.0000

Critical Finding: The single-device altitude-pressure correlation is $r = +0.0465$ (essentially zero), indicating that GNSS altitude variations are **not tracking actual atmospheric altitude changes**. This is a significant anomaly.

3.2.3 Per-Device Coupling Strength

Table 6: Altitude-Pressure Correlation by Device

Device ID	Correlation (r)	Records	Assessment
...948226	-0.022	18,462	Weak (Anomaly)
...527426	+0.023	14,185	Positive (Anomaly)
...508217	+0.033	15,971	Positive (Anomaly)
...250224	+0.046	19,682	Positive (Anomaly)
...437779	+0.058	19,039	Positive (Anomaly)
...605977	+0.096	16,925	Positive (Anomaly)
...369164	+0.104	12,913	Positive (Anomaly)
...373510	+0.331	17,991	Strong Positive (Severe)

Summary:

- Strong coupling devices ($r < -0.5$): **0**
- Weak coupling devices ($r \geq -0.3$): **8 (100%)**

3.3 Visualization

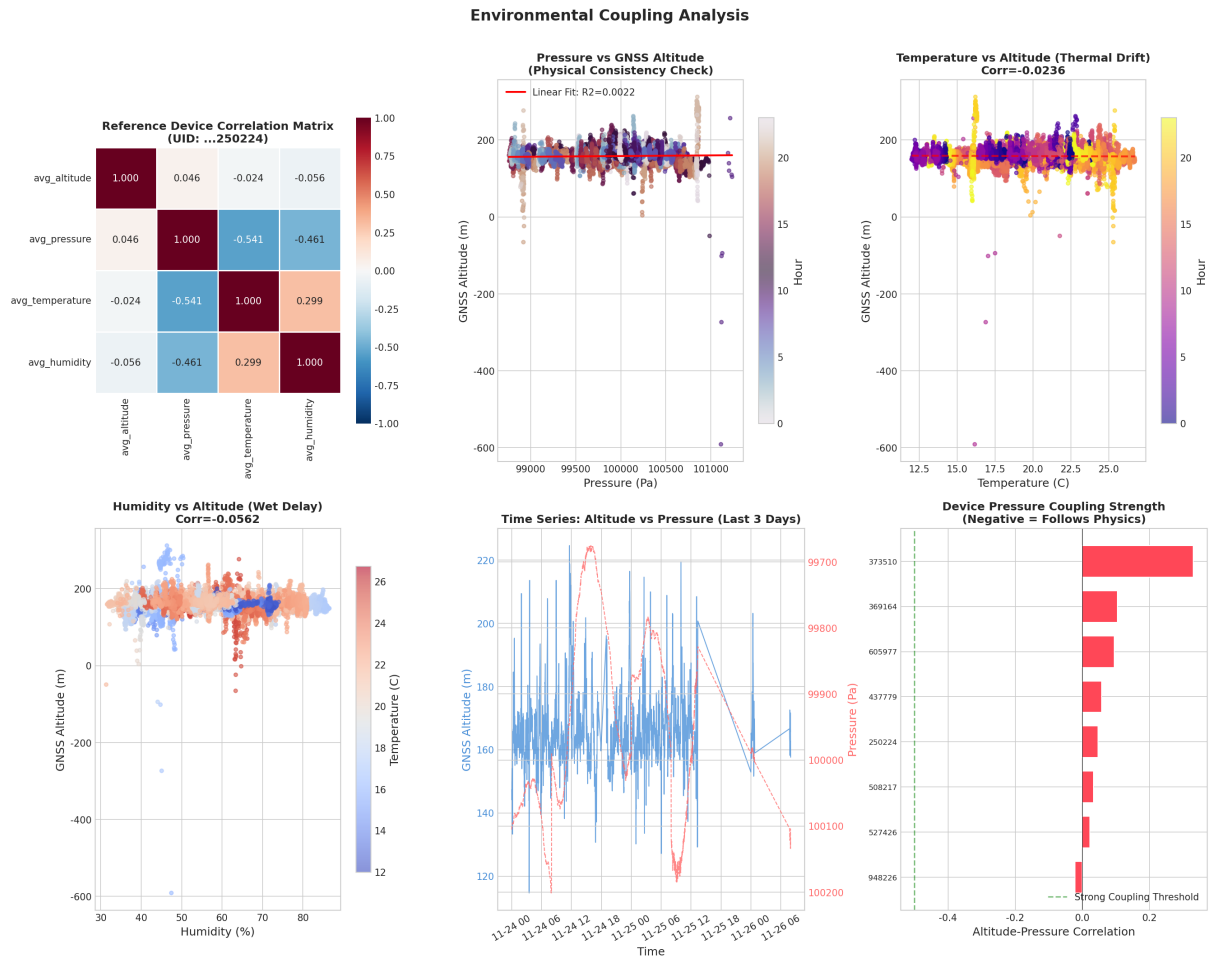


Figure 2: Environmental Coupling Analysis. Top row: Correlation heatmap, pressure vs altitude scatter with linear fit, temperature vs altitude scatter. Bottom row: Humidity vs altitude, time series comparison, per-device coupling strength.

3.4 Interpretation

The lack of expected negative altitude-pressure correlation within individual devices suggests:

1. **GNSS altitude noise dominates:** Random measurement errors are larger than actual altitude variations
2. **Static deployment:** Devices are stationary, so true altitude doesn't vary with weather
3. **Sensor decoupling:** The barometric sensor and GNSS receiver may not be physically co-located or time-synchronized

4 Dimension 3: Diurnal Cycle Analysis (Multipath Detection)

4.1 Theoretical Background

Multipath interference occurs when GNSS signals reflect off nearby surfaces (buildings, metal structures, water) before reaching the antenna. This manifests as:

- **24-hour periodicity:** As satellites traverse the sky, reflection geometry changes systematically
- **12-hour sub-harmonic:** Secondary peaks due to GPS orbital period (≈ 12 hours)
- **Amplitude $> 2\text{m}$:** Significant multipath typically produces meter-level errors

4.2 Results

4.2.1 Diurnal Amplitude Comparison

Table 7: Diurnal Amplitude: Stable vs Unstable Device

Device Category	Device ID	Altitude STD (m)	Diurnal Amplitude (m)
Most Stable	...605977	5.01	11.17
Least Stable	...250224	18.24	21.68

4.2.2 Multipath Risk Assessment

Table 8: Multipath Risk Distribution

Risk Level	Amplitude Threshold	Devices	Percentage
High Risk	$> 5\text{m}$	8	100%
Medium Risk	$2\text{--}5\text{m}$	0	0%
Low Risk	$< 2\text{m}$	0	0%

Critical Finding: All devices exhibit high-risk multipath signatures with diurnal amplitudes exceeding 5m .

4.3 Visualization

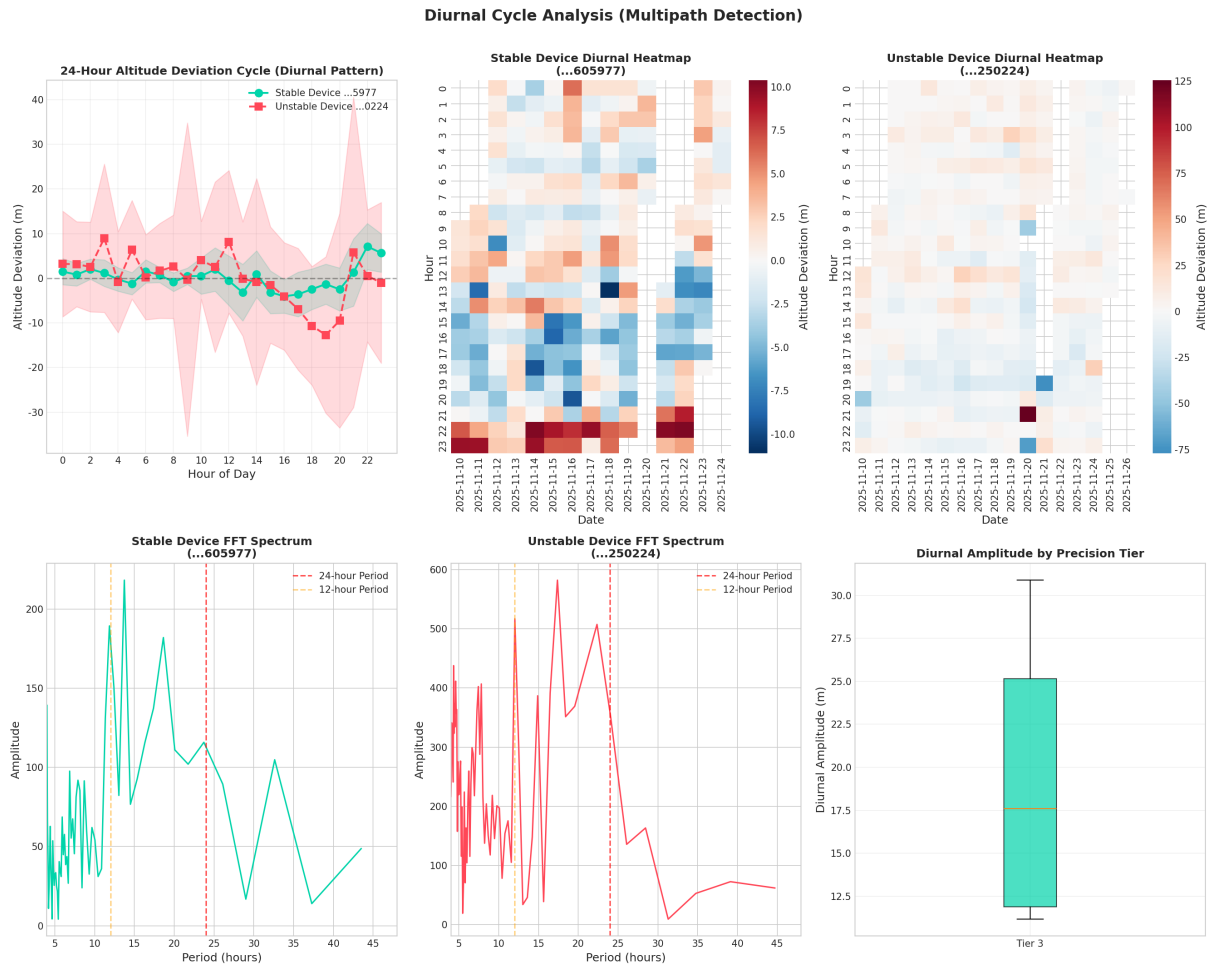


Figure 3: Diurnal Cycle Analysis. Top row: 24-hour altitude deviation pattern comparison, stable device heatmap, unstable device heatmap. Bottom row: FFT spectrum (stable), FFT spectrum (unstable), amplitude distribution by tier.

4.4 Interpretation

1. **Systematic 24-hour patterns** are evident in both stable and unstable devices, confirming multipath as a primary error source
2. **FFT analysis** reveals strong spectral components at 24-hour and 12-hour periods, consistent with GPS satellite orbital characteristics
3. **Even the “most stable” device** shows 11.17m diurnal amplitude, indicating that multipath affects the entire deployment
4. **Heatmap patterns** show consistent hour-by-hour deviations across multiple days, ruling out random noise

4.5 Recommended Actions

- Conduct site survey to identify reflective surfaces (metal roofs, glass facades, water features)
- Consider antenna relocation or installation of ground planes / choke rings

- Implement sidereal filtering to mitigate repeating multipath signatures
- Evaluate multi-frequency GNSS receivers for improved multipath rejection

5 Conclusions

5.1 Summary of Findings

1. Vertical Precision:

- 100% of devices classified as Tier 3 (Warning)
- Best device: STD = 5.01m; Worst device: STD = 18.24m
- One device shows 903m altitude range anomaly requiring investigation

2. Environmental Coupling:

- All devices show weak or anomalous altitude-pressure correlation
- No devices demonstrate expected negative coupling ($r < -0.5$)
- GNSS altitude variations appear decoupled from atmospheric physics

3. Multipath Effects:

- 100% of devices exhibit high-risk diurnal patterns (amplitude > 5 m)
- Strong 24-hour and 12-hour periodicities detected via FFT
- Systematic multipath interference confirmed across entire deployment

5.2 Data Quality Assessment

Table 9: Overall Data Quality Score

Metric	Score	Rating
Vertical Precision	0/8 pass Tier 2	Poor
Environmental Coupling	0/8 strong coupling	Poor
Multipath Resistance	0/8 low risk	Poor
Overall		Poor

5.3 Recommendations

1. **Immediate:** Inspect device 1878250224 for hardware issues
2. **Short-term:** Survey all antenna installation sites for multipath sources
3. **Medium-term:** Implement RTK/PPP corrections or relocate antennas
4. **Long-term:** Consider upgrading to multi-frequency GNSS with better multipath mitigation

A Appendix: Data Files

The following output files are available in the `data/reports/deep_analysis/` directory:

File	Description
<code>device_tier_stats.csv</code>	Complete device statistics with tier classification
<code>device_coupling_analysis.csv</code>	Per-device environmental coupling coefficients
<code>diurnal_amplitude_analysis.csv</code>	Diurnal amplitude measurements by device
<code>01_vertical_precision_tiering.png</code>	Precision tiering visualization
<code>02_environmental_coupling.png</code>	Environmental coupling analysis plots
<code>03_diurnal_cycle_multipath.png</code>	Diurnal cycle and multipath analysis

B Appendix: Methodology Notes

B.1 Tier Classification

Devices are classified based on the standard deviation of their altitude measurements over the observation period. Thresholds are based on typical GNSS performance expectations:

- Tier 1 ($\leq 1.5\text{m}$): Achievable with RTK/PPK or high-quality single-point positioning
- Tier 2 ($\leq 5.0\text{m}$): Standard single-point GNSS performance
- Tier 3 ($> 5.0\text{m}$): Indicates environmental issues or hardware problems

B.2 Correlation Analysis

Pearson correlation coefficients are computed between altitude and environmental factors. Strong negative altitude-pressure correlation is expected based on the barometric altitude formula.

B.3 FFT Analysis

Fast Fourier Transform is applied to detrended hourly altitude data to identify periodic components. A 24-hour rolling mean is subtracted to remove long-term trends before spectral analysis.